

Nathan Cook

Ms. Harper

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Automation in Enrichment: Is It Worth Implementing?

Albert Schweitzer, receiver of the 1952 Nobel Peace prize, once proclaimed that “Until he extends the circle of his compassion to all living things, man will not himself find peace.” Every National Zoo has deemed this principle priority zero, giving enrichment of all kinds to a plethora of species under their care for not only the happiness of guests, but of the animals themselves. Now, thanks to the ever-advancing achievements of technology, we have found a more interactive way of doing so. Zoos from all over the country have begun implementation of automated, or robotic enrichment, such as using touchscreens, motion sensors, and have even begun placing AI into the mix as a new method of inducing stimuli into a species (Scheer et al. 1). While the novelty of automated, or technological enrichment can sustain, and rotate the curiosity of a species more effectively than traditional enrichment could, furthermore reducing habituation rates, it is beneficial that we consider possible disadvantages with automation completely overtaking this field.

Specialists have been astounded to notice that this new enrichment has been not only effective in delivering proper care, and stimuli into species alike, but has passed its predecessors. Species must be stimulative in order to be mentally stable. Naturally, the variants of said species differentiate from one another. Automation, however, does a proper job at rotating frequencies, or adapting to fit the animal’s desired preferences, compared to previous enrichment, which

becomes customary over time, and sometimes will not fit a variation of a species. This is further exemplified through an experiment performed with American Mink. The Mink were put in a cage, and were tested by their reactions to enrichment items, and “Item exchange appeared to have a sustaining effect on exploration of all (Mink) in the cage, including familiar objects” (Clark et al.). Automation would allow for much more immersive, and reactive responses because of how animals respond to unfamiliar activity. It would adapt to the frequency of the species it attempts to stimulate, therefore becoming more engaging, and impactful, which is a struggle to apply to more traditional enrichment. This new method will provide less difficulty to specialists, and would ultimately accumulate savings the Zoo could use for more supplementary purchases (Kresnye et al. 3).

This new form of enrichment is looking to significantly downsize habituation from most species. When an animal is unsatisfied, or accustomed with its current enrichment, it becomes habitual, or disconnects from said enrichment. In laymen’s terms, it gets bored with the activities it’s given. While rotation is effective to decrease rates of habituation, many attributes could factor into the animal’s ability to stay “invested”. Habitat, sensory aspects, and dietary needs can all be contributed (Riley). This new enrichment is looking to minimize those issues by incorporating it into the animal’s environment, for example, creating an “artificial tree” with a touch screen to occupy a troop of apes. “The versatility and programmability of computer tasks makes them an ideal platform for achieving functionally naturalistic outcomes for great apes” (Scheerat al.). If zoos could implement automation into these quintessential elements of an animal's ability to generate stimuli, habituation can be effectively extinguished.

While there are a variety of benefits to this new approach, drawbacks are certainly apparent. Cost issues, species navigation (Whether this tech appeals to many, or all species), and applying to animal senses rather than human senses are all causes for concern (Kresyne et al.). The most prominent, and raised issue, however, is whether this new enrichment is in line with the motives of National Zoos. The morality of such enrichment, some said, goes against the Zoo's pursuit of public learning. "Is it really teaching the public the right ideas about wildlife to show orangutans playing 'video games'?" (Scheer et al. 6). Others have noticed that, just as with humans, it could be imprudent, damaging, even, for animals to go through extended periods interacting with something like a touch screen (Scheer et al. 1). These are both reasonable, and probable scenarios, and they should be further discussed. However, this depends on whether we completely discard traditional enrichment methods. Whilst automation is an amazing progression in this field, National Zoos should blend both old, and new. "Traditional enrichment techniques, such as puzzle feeders, novel objects, sensory stimuli, and food variation, have shown benefits including reduced stress and improved health." (Kresnye et al.). A well balanced mix is both physically, and mentally healthy for an animal. It might become even more stimulating, as one is by a human, and another by AI, or a digital appliance.

This new form of enrichment can flip the methodology of these National Zoos on its head, causing not only a better grasp of the stimulation of a species, but a better environment for them. Using rotation of different protocols in different areas of the animal's surroundings, the selected animal is kept naturally healthy, and invested. However, leaning too far to one side is unwise. It is crucial that we make use of all methods of enrichment, possibly even improving

upon the old with the conceptions we've learned from the new. In order to find peace, we must first discuss, compromise, and achieve proper balance.

Works Cited

- Clark, Gabrielle B, et al. "Diversity and Novelty in Environmental Enrichment Increases Enrichment Use in Juvenile American Mink (*Neogale vison*).” *Frontiers in Animal Science*, vol. 4, 9 Aug. 2023, <https://doi.org/10.3389/fanim.2023.1228533>.
- Kresnye, K. Cassie, et al. "Survey on the Past Decade of Technology in Animal Enrichment: A Scoping Review.” *Animals*, vol. 12, no. 14, 13 July 2022, p. 1792, <https://doi.org/10.3390/ani12141792>.
- Riley, Alex. Email to CART students. 4 Dec. 2025.
- Scheer, Benjamin, et al. "Technology-Based Cognitive Enrichment for Animals in Zoos: A Case Study and Lessons Learned.” *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 41, no. 0, 2019, escholarship.org/uc/item/6xq790f1. Accessed 3 Dec. 2025.